



INNOVATION/ ENTREPRENEURSHIP/SOCIETAL

Internship report

Submitted by

**VIKRAM PRAKASH
(4NI22EE046)**

*In partial fulfilment of the requirement
for the award of degree of Bachelor of Engineering
in*

Electrical and Electronics Engineering

Under the supervision of

Mrs. Jahnavi B P

Assistant Professor

Dept.of Electrical and Electronics Engineering
The National Institute of Engineering, Mysuru



**Department of Electrical & Electronics Engineering
The National Institute of Engineering**
(An Autonomous Institute under VTU, Belagavi)
Mysuru-570 008

April 2026

DEPARTMENT OF ELECTRICAL & ELECTRONICS ENGINEERING

THE NATIONAL INSTITUTE OF ENGINEERING

(An Autonomous Institution)

Manandavadi Road, Mysuru- 570 008

Phone: 0821-2480475, 2481220, 4004900 Fax: 0821-2485802, E-mail: principal@nie.ac.in

Website: www.nie.ac.in

CERTIFICATE

This is to certify that Mr. Vikram Prakash bearing USN: 4N122EE046 has undergone Internship (BEE803) at Mindset IT Solutions Mysuru. The Candidate has submitted the report on Internship after completion of the training in partial fulfilment of curriculum prescribed for the Sixth semester of B.E course in Electrical and Electronics Engineering during the year 2025-26.

Signature of Supervisor

Signature of HOD

Examiners

Signature with date

- 1.
- 2.
- 3.
- 4.

A Grant-in-Aid Institution Under Govt. of Karnataka
Permanently affiliated to Visvesvaraya Technology University, Belagavi.
Recognized by AICTE, New Delhi & Accredited by National Board of Accreditation
New Delhi.

CERTIFICATE



Date: 27th April 2026

To,
The Department of Electrical and Electronics Engineering,
The National Institute of Engineering (NIE)
,Mysuru.

Subject: Confirmation of Internship Status and Academic Progress

Respected Sir/Madam,

This is to formally confirm that the following students from **The National Institute of Engineering, Mysuru**, are currently engaged in an internship program at **MINDSET IT SOLUTIONS**, Bangalore. The internship focuses on **Embedded Mechatronics and IoT** and is scheduled from **25/01/2026 to 25/05/2026**.

The students listed below have been consistently complying with our organizational policies and management systems while working on their respective modules:

Student Details:

1. Sanjana P	15. Shreya Ashok Bilagi
2. Vikram Prakash	16. Shamanth R
3. Srujan V	17. Mahavardhan
4. Unnathi M N	18. Sunil M L
5. Sevanthi V	19. Hemavathi G
6. Ramkishan	20. Vidhya Raj
7. Hemanth T S	21. Shaun Mario F
8. K Prashanth	22. Manjunath N P
9. V Ganesha	23. J Manoj Gowda
10. Anagha U	24. Ashwini A
11. Adithi K	25. Srinivasa Ambekar
12. Pranathi G	26. Tarini V G
13. Dhanik Gowda K	27. Suchithra M
14. Sachin	

During this period, all ownership and usage rights of the work completed remain with the company. We are satisfied with their technical progress and professional conduct thus far.

This statement is issued upon the request of the students for their academic records and evaluation at NIE. We wish them continued success in their final month of the program.

For **MINDSET IT SOLUTION,**


HARISH S
PROJECT MANAGER



ACKNOWLEDGEMENT

We express our sincere and profound gratitude to our esteemed Principal, Dr. B S Nagendra Prarashar, for graciously providing the necessary facilities and unwavering support that enabled the successful completion of this project.

We respectfully convey our heartfelt thanks to the HoD Dr. Ananthapadmanabha B R, for his invaluable guidance, constant encouragement, and insightful motivation throughout the course of this work.

We extend our deep sense of appreciation to our project guide, Mrs. Sonaxi B Raikar Assistant Professor, for his dedicated supervision, constructive suggestions, and continuous support, which played a pivotal role in the successful accomplishment of this project. We also wish to acknowledge our friends and classmates for their cooperation, encouragement, and assistance during challenging phases of the project, particularly in addressing technical aspects.

Finally, we express our deepest gratitude to our parents, whose constant encouragement, patience, and moral support have been a source of strength throughout our academic journey.

Vikram Prakash
4NI22EE046

ABSTRACT

This report presents a comprehensive overview of the Embedded Systems and IoT internship undertaken at MindsetCraft Labs Pvt. Ltd. The internship focused on developing practical knowledge and hands-on skills in the design, development, and implementation of IoT-based systems and embedded applications. The study begins with an introduction to the fundamentals of the Internet of Things (IoT), highlighting its role in connecting physical devices through sensors, software, and network technologies for real-time data collection and communication.

During the internship, various concepts such as microcontroller programming, sensor interfacing, communication protocols, and real-time system design were explored. Practical exposure was gained through working with development boards like NodeMCU, implementing projects involving motor control, automation, and data monitoring. The training emphasized integrating hardware and software components to build efficient, reliable, and scalable IoT solutions.

Additionally, the internship enhanced problem-solving abilities, technical proficiency, and understanding of industry-relevant tools and technologies. The experience provided valuable insights into real-world applications of embedded systems in domains such as home automation, industrial control, and smart technologies. Overall, the internship served as a strong foundation for future advancements in the field of embedded systems and

CONTENTS

CHAPTER 1: INTRODUCTION

- 1.1 Overview of Internship
- 1.2 Objectives of Internship
- 1.3 Relevance of Internship

CHAPTER 2: ABOUT THE COMPANY

- 2.1 Company/Organization Profile
- 2.2 Vision and mission

CHAPTER 3: WORK RESPONSIBILITIES DURING THE INTERNSHIP

- 3.1 Description of Assigned Task
- 3.2 Scope of Work
- 3.3 Constraints and Requirements

4. Work Done / Internship Activities (13 WEEKS)

Chapter 5: Results & Analysis

- 5.1 Observations
- 5.2 Performance Results

CHAPTER 6: SKILLS ACQUIRED DURING THE INTERNSHIP

- 6.1 Technical Skills
- 6.2 Applications

Conclusion

References

CHAPTER 1: INTRODUCTION

1.1 Overview of Internship

The internship was carried out at MindsetCraft Labs Pvt. Ltd. in the domain of Embedded Systems and Internet of Things (IoT). The primary objective of this internship was to gain practical exposure to microcontrollers, programming, and real-time system development.

During the internship, both fundamental and advanced concepts of Arduino programming and NodeMCU were studied and implemented. The training included hands-on experience with various sensors, actuators, and communication modules, enabling the development of real-time embedded applications.

Several practical experiments and mini-projects were successfully completed, such as LED control, obstacle detection, air quality monitoring, motor control, and IoT-based systems using platforms like Blynk. These activities helped in understanding the integration of hardware and software components.

Overall, the internship provided a strong foundation in embedded systems and IoT, enhancing technical knowledge and preparing for real-world engineering applications.

1.2 Objectives of Internship

The main objective of the internship was to gain hands-on experience and practical knowledge in Embedded Systems and IoT, thereby bridging the gap between theoretical learning and real-time implementation.

The specific objectives are summarized below:

- To understand the fundamentals of Arduino programming, including variables, operators, and control structures
- To interface various sensors and devices such as IR sensors, LDR, DHT11, LCD displays, motors, and buzzers
- To develop IoT-based applications using NodeMCU
- To implement remote monitoring and control systems using platforms like Blynk
- To enhance problem-solving and analytical skills through practical applications
- To design and develop real-time projects such as obstacle detection, fire detection, and smart irrigation systems

This internship aimed to build a solid technical base and improve practical engineering skills required in industry.

1.3 Relevance of Internship

The internship holds significant relevance in today's technology-driven world, where Embedded Systems and IoT play a vital role in automation and smart solutions. It provides practical insight into how theoretical concepts are applied in real-world systems.

Through this internship, knowledge was gained in designing and developing systems capable of sensing, processing, and transmitting data. Exposure to microcontrollers, sensors, and IoT platforms helped in understanding modern applications used in domains such as smart homes, healthcare, agriculture, and industrial automation.

Furthermore, the internship enhanced technical competency, problem-solving ability, and hands-on experience, which are essential for a successful career in core engineering fields. It also aligned academic learning with industry requirements by introducing current technologies and real-time applications.

In conclusion, the internship is highly relevant as it strengthens the foundation for future advancements and career opportunities in Embedded Systems and IoT.

CHAPTER 2 - About MINDSETCRAFT LABS PVT LTD



DIRECTORS: MANJUNATH REDDY C S & SHRIDHAR PATIL

Address: #614/1 Behind Bajaj Showroom, KR Mohalla, Sunnadakeri, Mysore-570004

Phone: 9980975953 , 7338581758

Email: mindsetcraftlabs@gmail.com

Website: <http://www.mindsetmysore.in/>

2.1 About Us:

Mindset Craft Labs Pvt Ltd is a pioneering force in the realm of technology, dedicated to crafting innovative software solutions and nurturing the next generation of tech talent. Established by Shridhar Patil and Manjunath Reddy, our company stands at the intersection of creativity, expertise, and cutting-edge technology. We are also proud to be registered under MSME and Startup India, further strengthening our commitment to innovation and entrepreneurship.

We have won numerous Best Project and Innovation Awards in various competitions, showcasing our excellence in technological advancements. Our expertise extends to real-time applications, including both software and hardware-related services and products that cater to various industries and practical use cases.

2.2 Vision and Mission

Vision:

To become a leading organization in providing quality technical education and training in Embedded Systems and Internet of Things (IoT), empowering students with the skills required to excel in modern technology-driven industries.

Mission:

To deliver industry-oriented training through practical learning and real-time project development. The organization aims to enhance students' technical knowledge, problem-solving abilities, and innovation skills by providing hands-on experience with the latest tools and technologies in embedded systems and IoT.

Our Services:

Software and Hardware Projects:

Leveraging our extensive experience and expertise, we undertake a diverse range of software projects, including:

- Website Designing
- ERP Applications Development
- Machine Learning Applications Development
- Deep Learning Applications Development
- Embedded Projects Development
- IoT Projects Development
- Robotics Development
- Home Automations, etc.

CHAPTER 3: WORK RESPONSIBILITIES DURING THE INTERNSHIP

3.1 Description of Assigned Task

During the internship, various tasks were assigned to develop a strong understanding of Embedded Systems and IoT. The initial tasks included learning Arduino programming concepts such as variables, data types, operators, and control structures like if-else statements, loops, and switch cases. These concepts were applied through basic programs like LED blinking and fading.

Further tasks involved interfacing different sensors and devices with microcontrollers. This included working with IR sensors for obstacle detection, LDR for light sensing, DHT11 for temperature and humidity measurement, and LCD displays for data visualization. Practical implementations such as buzzer control, motor operation, and sensor-based automation systems were also performed.

Advanced tasks included developing real-time projects like fire detection systems, plant irrigation systems, and traffic density monitoring. Additionally, IoT-based applications were created using NodeMCU and Blynk for remote monitoring and control. These tasks helped in understanding the integration of hardware and software for building efficient and functional systems.

3.2 Scope of Work

The scope of work during the internship covered both theoretical understanding and practical implementation of Embedded Systems and IoT. It included learning and applying Arduino programming concepts to develop various real-time applications. The work primarily focused on interfacing microcontrollers such as Arduino and NodeMCU with different sensors and output devices.

The internship involved working with components like IR sensors, LDR, DHT11, LCD displays, motors, and buzzers to design and implement functional systems. It also included developing IoT-based applications using platforms like Blynk for remote monitoring and control.

Additionally, the scope extended to building and testing projects such as obstacle detection systems, fire detection systems, smart irrigation, and automation-based applications. Overall, the internship provided exposure to designing, coding, testing, and analyzing embedded and IoT systems in real-world scenarios.

3.3 Constraints and Requirements

During the internship, certain constraints and requirements were encountered while working on embedded systems and IoT-based projects. Proper hardware setup and accurate connections were essential for the correct functioning of the systems. Any mismatch in wiring or component configuration could lead to errors in output.

Reliable internet connectivity was required for IoT-based applications, especially when using platforms for remote monitoring and control. The performance of the system also depended on the accuracy and sensitivity of sensors such as IR, LDR, and DHT11.

Time limitations and availability of components posed minor challenges during project implementation. Additionally, debugging errors in both hardware and software required careful analysis and testing. Despite these constraints, all tasks were successfully completed by following proper procedures and systematic troubleshooting methods.

CHAPTER 4: Work Done / Internship Activities (13 WEEKS)

WEEK 1:

1. INTRODUCTION:

- The internet of things is the extension of internet connectivity into physical devices and everyday objects.
- IOT is simply the network of interconnected things/devices which are embedded with sensors, software, network, connectivity and necessary electronics that enables them to collect and exchange data making them responsive.
- It makes devices and systems taking all the things in the world and connecting them to the internet.
- The internet of things (IOT) refers to the ever-growing network of physical object that feature an IP address for internet connectivity, and the communication that occurs between these objects and others internet- enabled devices and systems.
- It is also referred to as Machine-to-Machine (M2M), Skynet or internet of everything.
- These devices collect useful data with the help of various existing technologies and then autonomously flow the data between other devices.



FIG 1.INTERNET OF THINGS

2. HISTORY:

The history of the Internet of Things (IoT) dates back to the early 1980s, when researchers began connecting everyday devices to the internet for automation and data sharing. One of the first examples was a network-connected Coca-Cola vending machine at Carnegie Mellon University that could report inventory and temperature. The term “IoT” was later coined in 1999 by Kevin Ashton, who envisioned a system where physical objects could communicate using sensors and RFID technology. In the 2000s, the growth of the internet, wireless communication, and embedded systems accelerated IoT development, and by the 2010s, it expanded rapidly with the rise of smartphones, cloud computing, and affordable

microcontrollers like ESP8266. Today, IoT is widely used in smart homes, healthcare, agriculture, and industrial automation, connecting billions of devices worldwide and transforming how humans interact with technology.

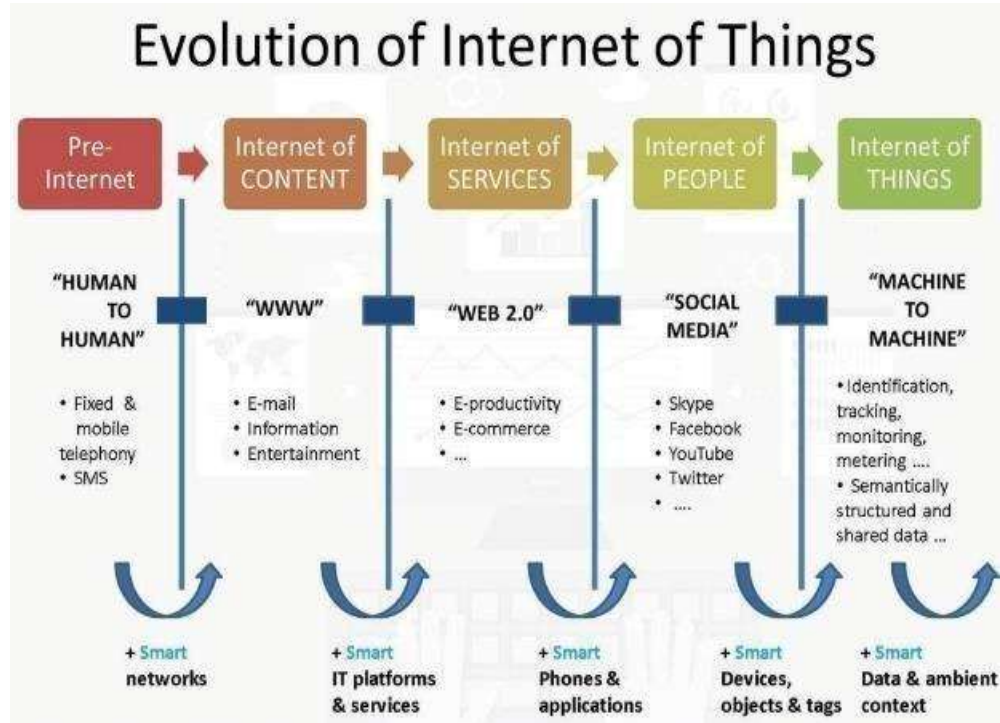


FIG 2. EVOLUTION OF IOT

3. LAYER OF INTERNET OF THINGS:

Internet of Things Reference Model

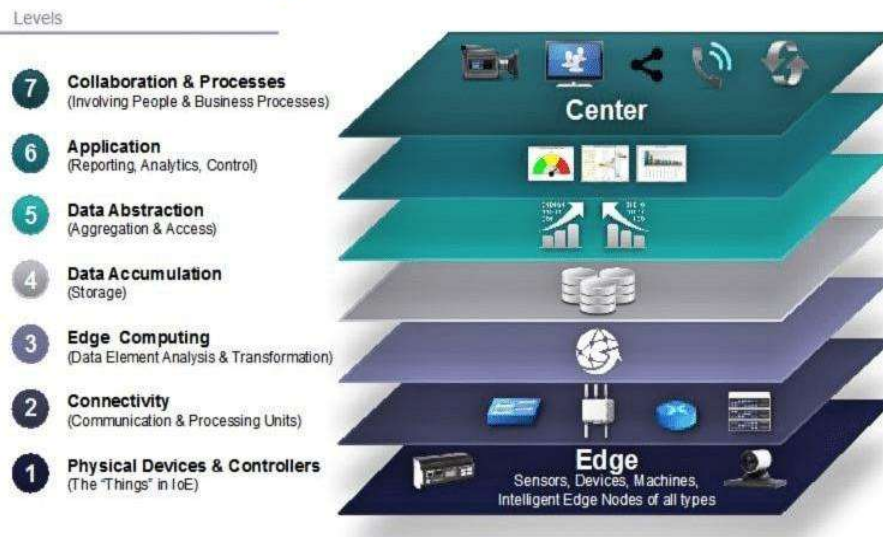


FIG 3. REFERENCE MODEL OF IOT

4. ARCHITECTURE OF IOT:

- The working of IOT is different for different IOT echo system (architecture). However, the key concept of there working are similar.
- The entire working process of IOT starts with the device themselves, such as smartphones, digital watches, electronic appliances, which securely communicate with the IOT platform.
- The platforms collect and analyze the data from all multiple devices and platforms and transfer the most valuable data with applications to devices.

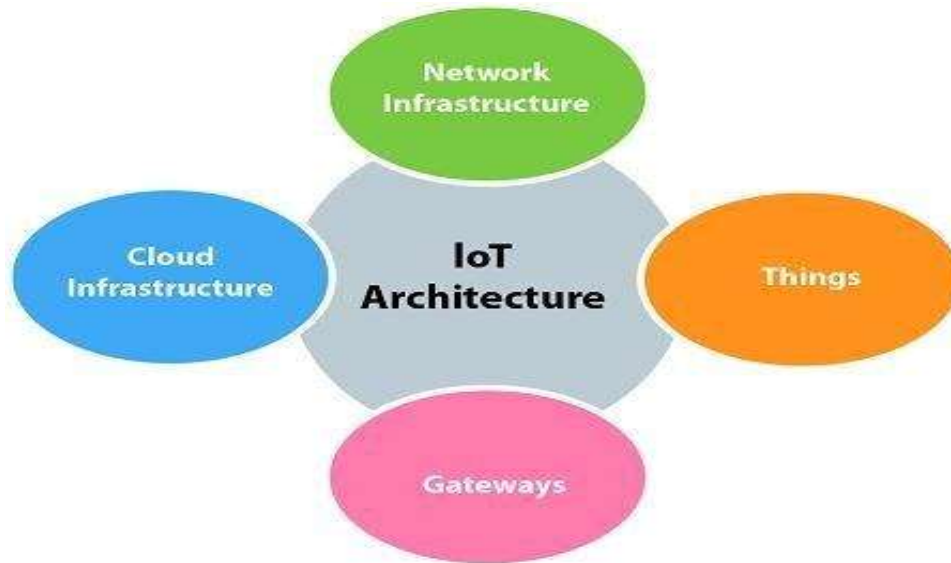


FIG 4. ARCHITECTURE OF IOT

WEEK 2:

5. COMPONENT OF IOT:

- Devices& sensors
- Connectivity
- Data processing
- User interface

1. Sensors:

A sensor is device that measures physical impact from its environment and converts it into that can be interpreted by a computer.

For example: speed, lights, sound and pressure.

2. Connectivity:

Several communications protocols and technologies are used in IoT to connect to internet cloud. Depending upon range, cost, power usage and the data rate etc. The right one used. For example: Bluetooth, mobile phones and wi-fi.

3. Data processing:

In the processing stage, a computer transforms the raw data into information. The transformation carried out by using different data manipulation techniques.

For example: collecting feed of all electrical applications and extracting car number from video feed of speeding cars.

4. User interface:

The information processed is made available to the end- user in some way, like giving alerts, notifications, monitoring, continuous feed or controlling the system remotely.

WEEK 3:

1.FEATURES OF IOT:

The most important features of IOT on which it works are connectivity, analyzing, integrating, active engagement, and many more. Some of them are listed below:

1.Connectivity: Connectivity refers to establish a proper connection between all the things of IOT to IOT platform it may be server or cloud. After connecting the IOT devices, it needs a high speed messaging between the devices and cloud to enable reliable, secure and bi-directional communication.

2.Analyzing: After connecting all the relevant things, it comes to real-time analyzing the data collected and use them to build effective business intelligence. If we have a good insight into data gathered from all these things, then we call our system has a smart system.

3. Integrating: IOT integrating the various models to improve the user experience as well.

4.Artificial Intelligence: IOT makes things smart and enhances life through the use of data. For example, if we have a coffee machine whose beans have going to end, then the coffee machine itself order the coffee beans of your choice from the retailer.

5.Sensing: The sensor devices used in IOT technologies detect and measure any change in the environment and report on their status. IOT technology brings passive networks to active networks. Without sensors, there could not hold an effective or true IOT environment.

6.Active Engagement: ActiveIOT makes the connected technology, product, or services to active engagement between each other.

7.Endpoint Management: It is important to be the endpoint management of all the IOT system otherwise, it makes the complete failure of the system. For example, if a coffee machine itself order the coffee beans when it goes to end but what happens when it orders the beans from a retailer and we are not present at home for a few days, it leads to the failure of the IOT system. So, there must be a need for endpoint management.

MICROPROCESSOR:

- A microprocessor is a computer processor where the data processing logic and control is included on a single integrated circuit, or a small number of integrated circuits.

- The microprocessor contains the arithmetic, logic, and control circuitry required to perform the functions of a computer's central processing unit.
- The integrated circuit is capable of interpreting and executing program instructions and performing arithmetic operations.
- The microprocessor is a multipurpose, clock-driven, register-based, digital integrated circuit that accepts binary data as input, processes it according to instructions stored in its memory, and provides results (also in binary form) as output.
- Microprocessors contain both combinational logic and sequential digital logic, and operate on numbers and symbols represented in the binary number system.
- The integration of a whole CPU onto a single or a few integrated circuits using Very-Large-Scale Integration (VLSI) greatly reduced the cost of processing power.
- Integrated circuit processors are produced in large numbers by highly automated metal-oxide- semiconductor (MOS) fabrication processes, resulting in a relatively low unit price.



FIG 10.1 MICROPROCESSOR

MICROCONTROLLER:

- A microcontroller (μ C or uC) is a solitary chip microcomputer fabricated from VLSI fabrication. A micro controller is also known as embedded controller.
- Today various types of microcontrollers are available in market with different word lengths such as 4bit, 8bit, 64bit and 128bit microcontrollers.
- Microcontroller is a compressed micro computer manufactured to control the functions of embedded systems in office machines, robots, home appliances, motor vehicles, and a number of other gadgets.
- A microcontroller is comprises components like - memory, peripherals and most importantly a processor. Microcontrollers are basically employed in devices that need a degree of control to be applied by the user of th



FIG 10.2 MICROCONTROLLER

What are embedded systems

Components of embedded system

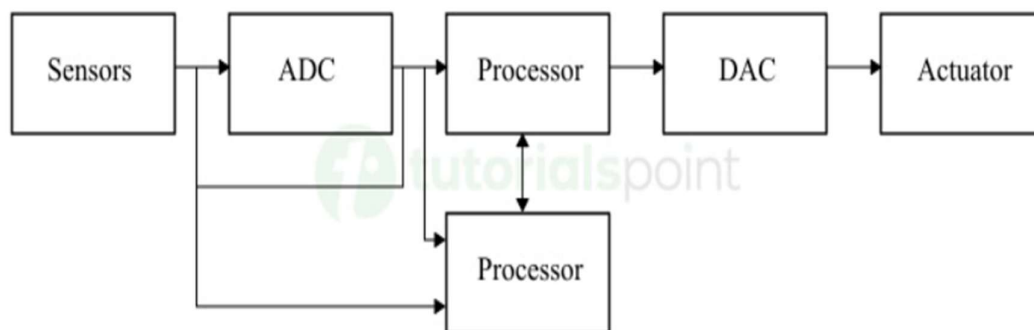
HARDWARE

SOFTWARE

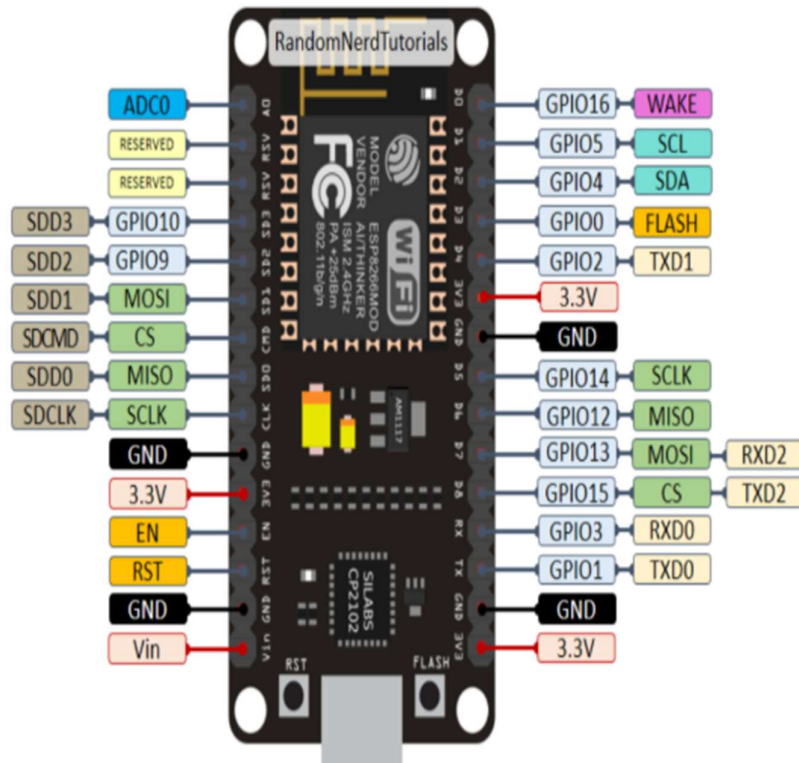
OPERATING SYSTEM

WEEK 4:

BLOCK DIAGRAM OF EMBEDDED SYSTEM



ARDUINO NODE MCU ESP8266



features of the NodeMCU ESP8266 board

NodeMCU ESP8266 – Features

1. Built-in Wi-Fi

- Uses ESP8266
- Supports:
 - 802.11 b/g/n
- Can work as:
 - **Station mode** (connect to Wi-Fi)
 - **Access Point mode** (create hotspot)

No external Wi-Fi module needed

2. Powerful Microcontroller

- 32-bit processor
- Clock speed: **80 MHz (can go up to 160 MHz)**
- Good for real-time IoT applications

3. GPIO Pins (Input/Output)

- Around **11–17 usable GPIO pins**
- Supports:
 - Digital input/output
 - PWM (for LED dimming, motor control)
 - I2C, SPI, UART communication

4. Internet Connectivity

- Directly connects to cloud platforms like:

- Blynk
- MQTT servers
- HTTP web servers

Perfect for IoT projects

5. Low Power Consumption

- Multiple power modes:
 - Active mode
 - Sleep mode
 - Deep sleep mode

Suitable for battery-powered devices

6. Easy Programming

- Can be programmed using:
 - Arduino IDE
 - Lua scripting
 - MicroPython

Beginner-friendly and widely supported

7. USB to Serial Interface

- Built-in USB (via CP2102/CH340 chip)
- No external programmer needed
- Easy code upload
-

8. Compact Size

- Small and lightweight
- Breadboard friendly

Ideal for portable projects

9. TCP/IP Stack Support

- Built-in network protocols:
 - TCP
 - UDP
 - HTTP

Enables web server/client applications

10. Security Features

- Supports:
 - WPA/WPA2 encryption
- Secure Wi-Fi communication

11. Analog Input

- 1 Analog pin (A0)
- Input range:

0–1V (internally scaled on board to 3.3V)

12. Flash Memory

- Typically **4MB flash**
- Stores:
 - Code

- Data
- Firmware

Why NodeMCU is Popular

- Low cost
- Built-in Wi-Fi
- Easy to use
- Huge community support

Common Applications

- Home automation
- Smart irrigation
- Weather monitoring
- IoT-based robotics
- Remote control systems

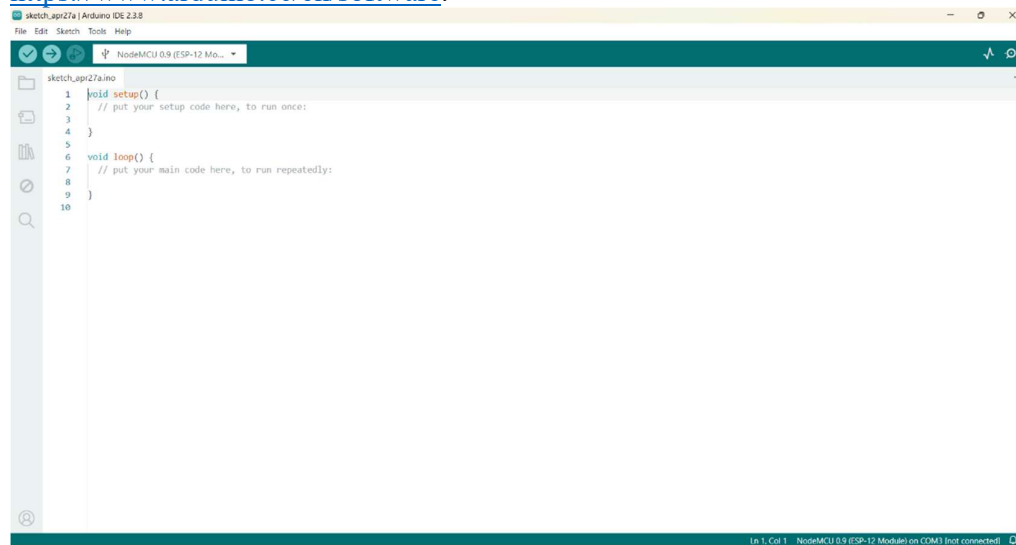
WEEK 5:

4. SOFTWARE DOWNLOAD:

- Arduino is a computation tool for sensing and controlling signals.
- It is more convenient and cost effective than using a personal computer pc.
- It's an open – source system in terms of hardware and software.

*You can download the integrated development environment (IDE) for your own OS form.

<https://www.arduino.cc/en/software>.



5. Tool bar details:

1. **Verify:** Checks code for errors
2. **Upload:** Compiles and uploads code to the Arduino i/o board.
3. **New:** Creates a new sketch.
4. **Open:** Open sketch.
5. **Save:** Save sketch
6. **Serial monitor:** Display serial data being sent from the arduino board

1. ARDUINO CODE PROGRAMMING BASIC CONCEPTS

DATA TYPES

The data types are used to identify the types of data and the associated functions for handling the data. It is used for declaring functions and variables, which determines the storage space.

- Char Data Type (1 byte)
- Int Data Type (2 bytes)
- Float Data Type (4 bytes)
- Double Data Type (4 bytes)
- String Data Type
- void Data Type

ARDUINO VARIABLES:

The variables are defined as the place to store the data and values. It consists of a name, value, and type.

The variables can belong to any data type.

Consider the below example

```
int s=3;
```

ARDUINO CONSTANTS

The constants are predefined expressions.

It makes the code easy to read.

Logical level Constants

Pin level Constants

ARDUINO OPERATORS

Arithmetic Operators

Compound Operators

Boolean Operators

Comparison Operators

WEEK 6:

Arduino control statements

If

If Else

Else if

Nested if

Arduino if statement

The if () statement is the conditional statement, which is the basis for all types of programming languages.

```
if ( condition)
```

```
{
```

```
// include statements
```

```
// if the condition is true
```

```
// then performs the function or task specified inside the curly braces
```

```
}
```

The if () statement is the conditional statement, which is the basis for all types of programming languages.

Arduino if else statement

The if-else condition includes if () statement and else () statement. The condition in the else statement is executed if the result of the If () statement is false

Arduino if else statement

The if-else condition includes if () statement and else () statement. The condition in the else statement is executed if the result of the If () statement is false

Arduino if , else if , else statement

The else if statement can be used with or without the else () statement. We can include multiple else if statements in a program

Arduino for loop

The statements inside the curly brackets under for loop are executed repeatedly according to the specified condition. An increment counter in the for loop is used to increment or decrement the loop repetitions.

Syntax:

```
for (initialization; condition; increment)
{
  \\ statements
}
```

initialization: It is defined as the initialization of the variable.

condition: The condition is tested on every execution. If the condition is true, it will execute the given task. The loop ends only when the condition becomes false.

increment: It includes the increment operator, such as i + +, i - - , i + 1, etc. It is incremented each time until the condition remains true

Arduino while loop

The while loop() is the conditional loop that continues to execute the code inside the parentheses until the specified condition becomes false.

The syntax is:

```
while (condition)
{
  // code or set of statements
}
```

Arduino do...while

The working of the do-while loop is similar to the while loop. The condition inside the do-while will execute at least once. It is because the condition is tested at the end of the loop instead of the beginning.

The syntax is:

```
do
{
// code or set of statements
} while (condition);
```

Arduino switch case

The switch case controls the flow of the program by executing the code in various cases. A switch statement compares a particular value of a variable with statements in other cases. When the statements in a case matches the value of a variable, the code associated with that case executes.

Syntax :

```
switch(variable)
{
case 1:
// statements related to case1
break;
case 2:
// statements related to case2
break;
case n:
// statements related to case n
break;
default:
// it contains the optional code
//if nothing matches in the above cases, the default statement runs
break;
}
```

WEEK 7:

Arduino Functions

The functions allow a programmer to divide a specific code into various sections, and each section performs a particular task. The functions are created to perform a task multiple times in a program.

Advantages of using Functions

- Let's discuss some advantages of using functions in programming, which are listed below:
- It increases the readability of the code.
- It conceives and organizes the program.
- It reduces the chances of errors.
- It makes the program compact and small.
- It avoids the repetition of the set of statements or codes.

Arduino input output

- Input pin mode

- Output pin mode
- ```
Void setup ()
{
pinMode (pin , mode);
}
```
- pin – the number of the pin whose mode you wish to set
  - mode – INPUT, OUTPUT, or INPUT\_PULLUP.

### **Arduino input output**

- digital Read(pin)
- digital Write(pin,value)
- analog Read(pin)
- Analog Write(pin,value)

### **Week 8:**

An I2C LCD display was interfaced with the microcontroller to display sensor data and messages. Programs for displaying static and scrolling text were developed. This improved understanding of communication protocols and display systems.

### **I2C LIQUIDCRYSTAL INTERFACE:**

```
#include<Wire.h> #include<LiquidCrystal_I2C.h> LiquidCrystal_I2C lcd(0X27,16,2);

void setup()
{

Serial.begin(9600);

lcd.init(); lcd.backlight(); lcd.clear(); delay(500);
Serial.println("the lcd programming"); lcd.setCursor(0,0);
lcd.print("NODE MCU INTERFACE");

lcd.setCursor(0,1);

lcd.print("WITH I2C LCD DISPLAY");
delay(5000); lcd.clear();
}

void loop()
```

```
{

;

}
```

### **Week 9:**

Air quality monitoring systems were developed using both analog and digital sensors. The system was designed to detect environmental pollution levels and trigger alerts based on threshold values, demonstrating real-world IoT applications.

### **Week 10:**

DC motor control was implemented using appropriate driver circuits. The motor was made to rotate in both clockwise and anticlockwise directions. This week helped in understanding motor control, switching, and interfacing techniques.

### **PROGRAM TO ROTATE DC MOTOR OPERATION IN CLOCKWISE AND ANTICLOCKWISE DIRECTION**

```
int pos=D0; int neg=D1; void setup()
{

 pinMode(pos,OUTPUT); pinMode(neg,OUTPUT); digitalWrite(pos,LOW);
 digitalWrite(neg,LOW);
}

void loop()
{

 digitalWrite(pos,HIGH); digitalWrite(neg,LOW); delay(5000);
 digitalWrite(pos,LOW); digitalWrite(neg,LOW); delay(2000); digitalWrite(pos,LOW);
 digitalWrite(neg,HIGH); delay(5000); digitalWrite(pos,LOW); digitalWrite(neg,LOW);
 delay(2000);
}
```

## **WEEK 11:**

On this day, I practiced implementing a combined sensor and output control program using a microcontroller. I worked with multiple components including an IR sensor, push button, LED, and buzzer. In the first program, I configured the IR sensor and button as inputs and the LED and buzzer as outputs using the `pinMode()` function. The button was configured using `INPUT_PULLUP` to ensure stable input readings. I wrote a program that reads the values from both the IR sensor and the push button using `digitalRead()`. Based on the sensor inputs, the microcontroller performs different actions. When the IR sensor detects an obstacle, the LED turns ON and a message is displayed on the serial monitor. When the push button is pressed, the buzzer is activated. If both the obstacle is detected and the button is pressed simultaneously, both the LED and buzzer blink together to indicate that both conditions are active. In addition to this, I completed an assignment program that simulates a smart street light system. In this program, I used an LDR sensor and an IR sensor along with an LED. The LDR sensor detects whether the environment is bright or dark, while the IR sensor detects the presence of an obstacle (such as a vehicle or person). When darkness and obstacle detection occur at the same time, the LED turns ON for a few seconds to simulate a street light automatically lighting up when someone passes by at night. If either condition is not met, the LED remains OFF. This helped me understand how multiple sensors can work together in an automation system.

## **WEEK 12:**

Communication modules such as GSM and Bluetooth were introduced. A GSM-based alert system was implemented to send notifications, and a Bluetooth-controlled robot was developed. This enhanced knowledge of wireless communication in embedded systems.

### **GSM INTERFACING CODE**

```
int ir=D0; void setup() {
 Serial.begin(9600); pinMode(ir,INPUT);
}

void loop() {

 int irv=digitalRead(ir); if(irv==LOW)
 {

 Serial.println("AT"); delay(1000); Serial.println("AT+CMGF=1");
 delay(1000); Serial.println("AT+CMGS=\"+91xxxxxxxxxx\"r"); delay(1500);
 Serial.println("INTRUDER DETECTED"); delay(1000);
 Serial.println((char)26); delay(10000);
 }

 else
```

```
{

 Serial.println("NO INTRUDER DETECTED");

}

}
```

## **WEEK 13:**

This week is focusing on project

### **Greenhouse Environmental Monitoring System Using Arduino**

#### **1. Introduction**

Agriculture has evolved with the integration of modern technologies that enhance productivity, efficiency, and sustainability. In greenhouse environments, maintaining optimal soil moisture is crucial for proper plant growth and nutrient absorption. Traditional methods of monitoring soil moisture are manual, time-consuming, and often inaccurate.

To overcome these challenges, an Arduino-based Greenhouse Environmental Monitoring System is developed. This system continuously monitors soil moisture levels using a sensor and provides real-time feedback. It helps in efficient irrigation management, reduces water wastage, and improves overall crop health.

#### **2. Objectives**

The primary objective of this project is to design and implement an automated soil moisture monitoring system using Arduino. The specific objectives are:

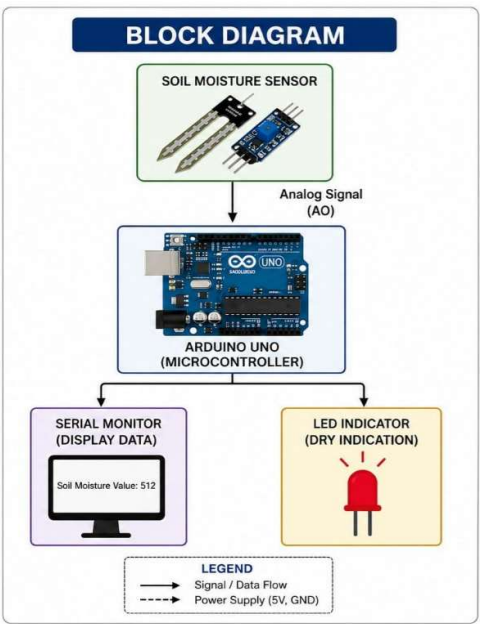
- To measure soil moisture continuously
- To convert soil conditions into electrical signals
- To process sensor data using a microcontroller
- To display real-time moisture values
- To provide visual indication using LED
- To reduce manual monitoring
- To improve irrigation efficiency

**3. System Overview**The system consists of three main components: a soil moisture sensor, an Arduino Uno microcontroller, and an LED indicator. The soil moisture sensor detects the water content in the soil based on electrical resistance. Dry soil offers high resistance, while wet soil offers low resistance.

The sensor converts this variation into an analog signal, which is read by the Arduino. The microcontroller processes the data and compares it with a predefined threshold value. Based on this comparison, the system determines whether the soil is dry or moist and controls the LED accordingly.

**4. Components and Specifications**

| Component            | Description           | Specifications                                    |
|----------------------|-----------------------|---------------------------------------------------|
| Arduino Uno          | Microcontroller board | 5V operating voltage, 16 MHz clock, 6 analog pins |
| Soil Moisture Sensor | Detects soil moisture | Analog (AO) and Digital (DO) outputs              |
| LED                  | Visual indicator      | Indicates dry/wet condition                       |
| Resistor             | Current limiter       | 220 Ω                                             |
| Breadboard           | Circuit assembly      | Solderless connections                            |
| Jumper Wires         | Connectivity          | Flexible wiring                                   |



## 5. Circuit Design

The system is divided into two parts: sensor input and output indication.

- Sensor Connections:  
VCC → 5V (Arduino)  
GND → GND (Arduino)  
AO → A0 (Arduino)
- LED Connections:  
Anode → Digital Pin 7  
Cathode → Resistor → GND

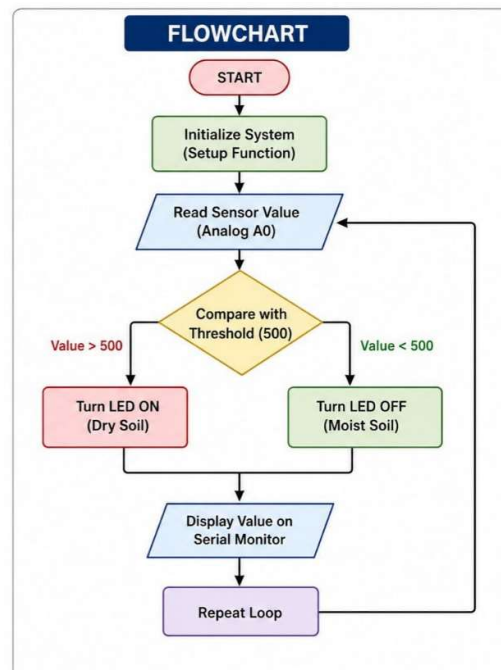
The sensor circuit captures soil moisture data, while the LED circuit provides visual feedback.

## 6. Working Principle

The soil moisture sensor measures the moisture level and generates an analog voltage signal. This signal is read by the Arduino through analog pin A0. The built-in ADC converts this signal into a digital value ranging from 0 to 1023.

The Arduino compares this value with a threshold. If the value exceeds the threshold, the soil is considered dry and the LED turns ON. If the value is below the threshold, the soil is moist and the LED remains OFF. The readings are continuously displayed on the Serial Monitor.

## 7. Software Implementation



The system is programmed using Arduino IDE. The program consists of two main functions:

- `setup()` function: Initializes serial communication and sets the LED pin as output.
- `loop()` function: Continuously reads sensor values, displays them, compares with threshold, and controls the LED.

## 8. Calibration

| Soil Condition    | Sensor Value Range |
|-------------------|--------------------|
| Wet Soil          | 0 – 400            |
| Moderate Moisture | 400 – 600          |
| Dry Soil          | 600 – 1023         |

The threshold value is typically set around 500. It can be adjusted based on soil type and environmental conditions for better accuracy.

### Code:

```
// Define pin connections
```

```
int sensorPin = A0; // Soil moisture sensor connected to A0
```

```
int ledPin = 7; // LED connected to digital pin 7
```

```
int sensorValue = 0; // Variable to store sensor value
```

```
int threshold = 500; // Threshold value for dry soil
```

```
void setup() {
```

```
 Serial.begin(9600); // Initialize serial communication
```

```
 pinMode(ledPin, OUTPUT); // Set LED pin as output
```

```
}
```

```
void loop() {
```

```
 // Read sensor value
```

```
 sensorValue = analogRead(sensorPin);
```

```
 // Print value to Serial Monitor
```

```
Serial.print("Soil Moisture Value: ");
Serial.println(sensorValue);

// Check condition
if (sensorValue > threshold) {
 digitalWrite(ledPin, HIGH); // Turn LED ON (Dry soil)
} else {
 digitalWrite(ledPin, LOW); // Turn LED OFF (Moist soil)
}

delay(1000); // Delay for 1 second
}
```

## **9. Advantages**

The system provides automated monitoring, real-time data, and efficient water usage. It is low-cost, easy to implement, and suitable for small-scale agricultural applications. It can also be expanded for advanced systems.

## **10. Applications**

This system can be used in greenhouse monitoring, smart irrigation systems, home gardening, and precision farming. It is especially useful where continuous monitoring is required.

## **11. Limitations**

The sensor may corrode over time due to continuous exposure to moisture. The accuracy depends on soil type and proper calibration. The system is basic and needs enhancements for large-scale use.

## **12. Conclusion**

The Greenhouse Environmental Monitoring System using Arduino is a simple and effective solution for monitoring soil moisture. It helps optimize irrigation, conserve water, and improve plant health. This project serves as a foundation for developing advanced smart agriculture systems.

## **CHAPTER 5: RESULTS & ANALYSIS**

### **5.1 Observations**

During the internship, several important observations were made while working with embedded systems and IoT-based applications. It was observed that sensors play a vital role in collecting real-time environmental data, which is then processed by the microcontroller to perform desired operations. The overall accuracy and efficiency of the system depend largely on proper sensor calibration and correct hardware connections.

It was also noted that IoT platforms significantly enhance system functionality by enabling remote monitoring and control. This makes the system more flexible, efficient, and user-friendly. Proper synchronization between hardware components and software programming is essential to ensure smooth and reliable system performance.

Furthermore, minor errors in coding or circuit connections can lead to incorrect outputs, emphasizing the importance of careful debugging and systematic testing. The experiments conducted during the internship demonstrated that embedded systems are highly reliable and effective when properly designed and implemented. These systems have wide applications in automation, smart devices, and modern technological solutions.

### **Key Results**

- All embedded system programs and IoT-based projects were successfully designed, implemented, and tested under various conditions.
- Sensor-based systems demonstrated high accuracy in detecting environmental parameters, including obstacle detection, fire detection, and air quality monitoring.
- Real-time monitoring and control were effectively achieved using NodeMCU integrated with IoT platforms.
- Hardware components such as LEDs, motors, buzzers, and LCD displays operated smoothly and reliably throughout the experiments.
- Automation systems, including smart irrigation and fire alert systems, functioned efficiently with minimal human intervention.
- Wireless communication using GSM and Bluetooth modules was found to be stable, responsive, and reliable.
- Overall system performance was consistent, efficient, and well-suited for real-time embedded and IoT applications.

### **5.2 Analysis**

The results obtained from various experiments and projects indicate that embedded systems and IoT technologies are highly effective for real-time monitoring and automation applications. The integration of sensors with microcontrollers enabled accurate data acquisition and timely response.

IoT-based implementations using NodeMCU allowed seamless remote access and control, improving system usability and efficiency. The performance of hardware components was consistent, and automation systems functioned reliably with minimal human intervention. Overall, the system performance was found to be stable and efficient. The

outcomes validate that embedded systems and IoT solutions are practical, scalable, and highly beneficial for modern engineering applications.

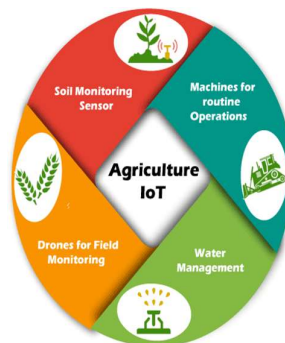
## **CHAPTER 6: SKILLS ACQUIRED DURING THE INTERNSHIP**

### **6.1APPLICATION OF IOT:**

1. Agriculture
2. Smart home
3. Smart cities
4. Health care
5. Connected cars
6. Machine 2 machine

#### **1. AGRICULTURE:**

- Field management – strategically placed IOT sensors will enable collection of crucial data such as field history, soil moisture , and vegetation map.
- Climate analysis& forecast – installing IOT – powered weather stations will allow monitoring climate conditions and help forecast weather.
- Equipment tracking - IOT sensors installed will enable location tracking and predictive maintenance of machine by identifying defects.
- Energy & water conservation- IOT sensors can help monitor water consumption and energy consumption. Also identify faulty pipes &leaking equipment to save water.



## 2. SMART HOME:

- Device abstraction layer – the local interface that applications on the gateway use.
- Device driver interface – the interface to integrate HAN technologies.
- Remote interface – the interface between an operator cloud platform and the gateway.
- \*Cloud API – the interface provided to third parties from the operator cloud platform.



FIG5.2 SMART HOME

## 3. SMART CITIES:

- Resource management- collecting energy usage and water quality data to analyze and optimize consumption.
- Waste management – introducing smart recycling systems, improving sanitation.
- \*Eco friendly infrastructure- analyzing and reducing waste , mindful consumption natural resources,vehicle sharing system.
- Smart lighting- dimming street lights when no one around to save energy.



FIG 5.3 SMART CITY

#### 4. HEALTH CARE:

- Telemedicine - remote appointment, remote diagnostics, remote monitoring, medication adherence.
- Hospital environmental condition monitoring- air quality, temperature , hazard sensing, lighting control.
- Connected ambulance - remote diagnostics, remote communication and data transfer.
- Immersive media – media training simulation with VR, AR decision support testing.

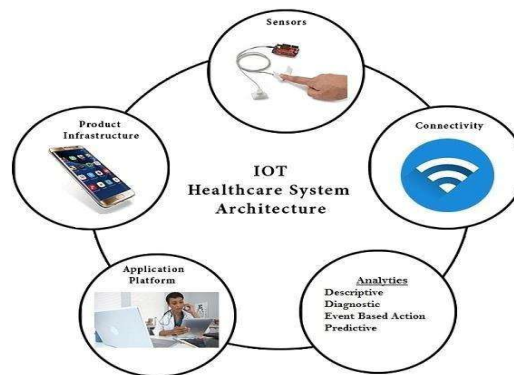


FIG5.4 HEALTH CARE OF IOT

#### 5. CONNECTED CARS:

- The cars use several sensors and embedded system connected to the

cloud and the internet to keep generating data and sending them to the cloud for informed decision –making through machine learning.

- Through it will take a few more years for the technology to evolve completely and for countries to amend laws and policies, what we're witnessing right now is one of the best application of IOT.

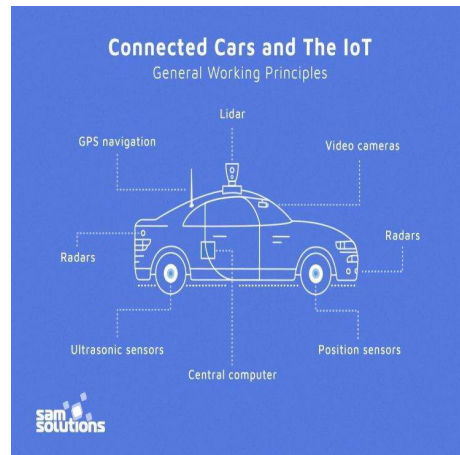


FIG 5.5 CONNECTED CARS

## 6. MACHINE 2 MACHINE:

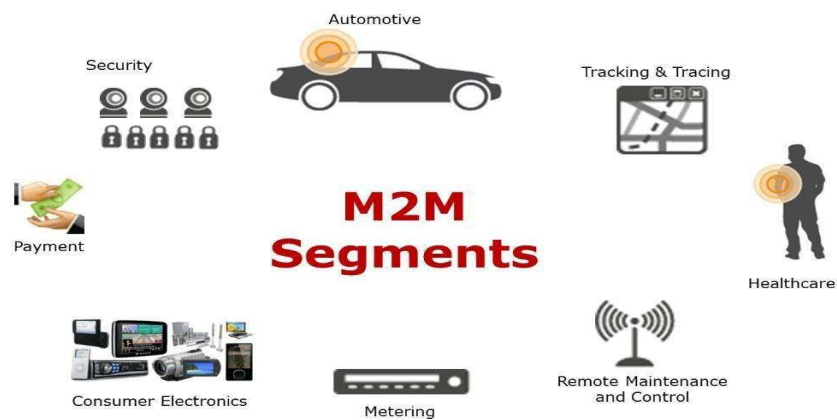


FIG 5.6 MACHINE 2 MACHINE

## Conclusion

The internship at **MindsetCraft Labs Pvt. Ltd.** provided a comprehensive and valuable learning experience in the field of Embedded Systems and Internet of Things (IoT). It effectively bridged the gap between theoretical knowledge acquired during academic studies and its practical application in real-world scenarios.

During the internship, both fundamental and advanced concepts—including Arduino programming, microcontroller operations, sensor interfacing, and IoT communication—were thoroughly understood and successfully implemented. A variety of experiments and projects, such as LED control, obstacle detection, environmental monitoring, motor control, fire detection, smart irrigation, and IoT-based remote monitoring systems, were carried out. These activities enhanced the understanding of hardware–software integration for developing efficient and automated systems.

The use of platforms like NodeMCU and Blynk further strengthened knowledge in wireless communication and remote device control. Additionally, the internship significantly contributed to the development of practical and analytical skills, including circuit design, debugging, system analysis, and problem-solving.

Apart from technical knowledge, important professional skills such as time management, teamwork, adaptability, and technical documentation were also improved.

In conclusion, the internship established a strong foundation for future growth in Embedded Systems and IoT. It enhanced technical competence, improved confidence in handling real-time engineering challenges, and will be highly beneficial for higher studies and career opportunities in modern technology-driven industries.

## References

- [1] Arduino, “Arduino Official Website – Microcontroller Programming and Development,” [Online]. Available: <https://www.arduino.cc>. [Accessed: Apr. 28, 2026].
- [2] ESP8266, “NodeMCU Documentation – IoT Development Platform,” [Online]. Available: <https://nodemcu.readthedocs.io>. [Accessed: Apr. 28, 2026].
- [3] Blynk Inc., “Blynk IoT Platform – Remote Monitoring and Control,” [Online]. Available: <https://blynk.io>. [Accessed: Apr. 28, 2026].
- [4] DHT11 Sensor, “Temperature and Humidity Sensor Datasheet,” [Online]. Available: Manufacturer Datasheet. [Accessed: Apr. 28, 2026].
- [5] IR Sensor Module, “Obstacle Detection Sensor Documentation,” [Online]. Available:

Technical Documentation. [Accessed: Apr. 28, 2026].

[6] LDR Sensor, “Light Dependent Resistor – Working and Applications,” [Online]. Available: Technical Resources. [Accessed: Apr. 28, 2026].

[7] I2C LCD Module, “LCD Interfacing using I2C Protocol,” [Online]. Available: Technical Guide. [Accessed: Apr. 28, 2026].

[8] GSM Module, “GSM Communication Module Documentation,” [Online]. Available: Technical Documentation. [Accessed: Apr. 28, 2026].

[9] Bluetooth Module, “HC-05/HC-06 Bluetooth Module Guide,” [Online]. Available: Technical Documentation. [Accessed: Apr. 28, 2026].

[10] MindsetCraft Labs Pvt. Ltd., “Embedded Systems and IoT Internship Training Materials,” [Online]. Available: Internal Resources. [Accessed: Apr. 28, 2026].